Wave Optics Question Bank

Part A – 2 Mark Questions (20)

1. Define wave optics.
2. State the principle of superposition.
3. What is interference of light?
4. Write the condition for constructive interference.
5. Write the condition for destructive interference.
6. What is a Michelson interferometer?
7. Define diffraction.
8. Differentiate between Fresnel and Fraunhofer diffraction.
9. What is Fraunhofer diffraction?
10. Write the condition for central maximum in single-slit diffraction.
11. Define grating element in a diffraction grating.
12. Write the grating equation.
13. What is polarization of light?
14. State Brewster's law.
15. Define double refraction.
16. Name the crystal used to demonstrate double refraction.
17. What is an analyzer?
18. What is a polarimeter?
19. Write two applications of polarization.
20. State Malus law.

Part B – 10 to 16 Mark Questions

1. Explain the principle of superposition and derive the expression for intensity in interference of light.
2. Describe the construction and working of Michelson interferometer with neat diagram.
3. Explain Fresnel's diffraction with example.
4. Explain Fraunhofer diffraction at a single slit and obtain the condition for minima and maxima.
5. Derive the expression for intensity distribution in a plane diffraction grating.
6. Compare interference and diffraction phenomena with examples.
7. Define polarization. Explain Brewster's law with neat diagram.
8. Explain double refraction in calcite crystal with ray diagram.
9. Explain the construction and working of a polarimeter with neat diagram.
10. Write short notes on applications of polarization in engineering and science.

Part C – Numerical Problems

1. Two coherent sources produce interference fringes with fringe width of 0.2 mm. If wavelength of light is 600 nm and distance between screen and sources is 2 m, find separation between sources.
2. Light of wavelength 589 nm is incident on a diffraction grating with 4000 lines/cm. Find the angle of second-order maximum.
3. In a single-slit diffraction, width of central maximum is 2 mm on a screen 2 m away. If wavelength is 600 nm, calculate the slit width.
4. Calculate the Brewster's angle for glass of refractive index 1.5.
5. A Michelson interferometer uses monochromatic light of wavelength 600 nm. If one mirror is moved by 0.3 mm, how many fringes will cross the field of view?
6. A double slit experiment is performed with slits separated by 0.2 mm and distance between slits and screen 1 m. If 10th bright fringe is at 2.5 cm, calculate wavelength of light.

7. Find the condition for missing orders in a diffraction grating where slit width is half of grating element.
8. Unpolarized light passes through two polaroids. If the transmission axis of the second makes 30° with the first, calculate the transmitted intensity.
9. A plane diffraction grating has 15000 lines per inch. Find the angular separation between two spectral lines of wavelength 600 nm and 602 nm in second order.
10. A polarimeter uses sodium light of wavelength 589 nm. If specific rotation of sugar solution is 66° per dm per gm/cc and length of tube is 20 cm, calculate concentration of solution when observed rotation is 13.2°.